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INVITED COMMENTARY

Confirmation Bias in Sport Science: Understanding and Mitigating Its Impact

Marco Beato,[1] Alexander T. Latinjak,[1] Maurizio Bertollo,[2] and Daniel Boullosa[3]

[1]School of Allied Health Sciences, University of Suffolk, Ipswich, United Kingdom; [2]Behavioral Imaging and Neural Dynamics Center, Department of Medicine and Aging Sciences, University "G. d'Annunzio" of Chieti-Pescara, Chieti, Italy; [3]Department of Physical Activity and Sports Sciences, Universidad de León, León, Spain

Purpose: Confirmation bias, a cognitive phenomenon deeply ingrained in human thinking, shapes our perception of reality. It leads individuals to favor information that aligns with their preexisting beliefs while dismissing or undervaluing the contrary evidence. In sport science, where precision and evidence-based practices are critical aspects to optimize performance, confirmation bias can decisively impact decision-making processes. In this context, practitioners rely on both objective and subjective data to be interpreted to enhance athletic performance. However, confirmation bias introduces a subtle yet powerful distortion, which is difficult to control and quantify. We usually believe that our observations can justify expectations or predictions about future observations, as well as broader generalizations beyond what we have directly observed—inductive inferences. Researchers and practitioners may therefore unknowingly seek out evidence that confirms their beliefs, inadvertently overlooking alternative viewpoints and interpretations. ***Conclusions:*** This paper explores the concept of confirmation bias in sport science, providing practical examples in the contexts of testing, training, and monitoring. By understanding how confirmation bias operates (ie, heuristic of decision making), we may develop strategies to mitigate its effects and promote more accurate and effective decision making in sport science (eg, performance).

Keywords: football, training, coaching, soccer

The discovery of instances which confirm a theory means very little if we have not tried, and failed, to discover refutations.

Karl Popper—*The Poverty of Historicism*[1]

Confirmation bias, a cognitive phenomenon deeply ingrained in human thinking,[2] shapes our perception of reality. It leads individuals to favor information that aligns with their preexisting beliefs while dismissing or undervaluing the contrary evidence. In sport science, where precision and evidence-based practices are critical aspects of optimizing performance, confirmation bias can decisively impact decision-making processes.[3] Sport science encompasses a diverse range of disciplines, ranging from exercise physiology and medicine to sport biomechanics and psychology. In this context, practitioners rely on both objective (eg, Global navigation satellite system)[4,5] and subjective (eg, rate of perceived exertion)[6–8] data to be interpreted to enhance athletic performance. However, confirmation bias introduces a subtle yet powerful distortion, which is difficult to control and quantify. We usually believe that our observations can justify expectations or predictions about future observations, as well as broader generalizations beyond what we have directly observed—inductive inferences.[9] Researchers and practitioners may therefore unknowingly seek out evidence that confirms their beliefs, inadvertently overlooking alternative viewpoints and interpretations. For instance, consider a strength and conditioning coach designing a training program for an elite athlete. If the coach harbors a belief that steady-state "cardio" is the ultimate performance enhancer, they may selectively focus on studies and athletes' feedback supporting this view. Meanwhile, research suggesting the benefits of other training modalities such as high-intensity interval training, resistance training, or their combination (ie, concurrent training) might receive less attention within this picture. Consequently, the athlete's training regimen becomes skewed, potentially limiting progress and further optimization. Similarly, we may consider the example of cycling coaches with some of them focused nowadays on the use of fine-tuning approaches of combining both power-meter-based metrics with metabolic and perceptual data.[10] In contrast, other coaches preferentially use power meters and ignore important metabolic or cognitive and affective data, which can limit the monitoring and evaluation of the physical, physiological, and psychological state of athletes. Another example, in sport psychology and motor learning domains, is the debate between the supporters of the ecological versus cognitive approach in which the researchers that take one of the 2 approaches tend to ignore the other.[11] Specifically, the ecological approach in sport psychology focuses on how athletes adapt their movements based on the interaction with their environment and task constraints, viewing learning as a dynamic process of exploration. In contrast, the cognitive approach emphasizes the mental processes involved in learning skills, breaking down movements into smaller parts and practicing them in isolation before integrating them into complex tasks.[12]

This paper explores the concept of confirmation bias in sports science, providing practical examples in the contexts of testing, training, and monitoring. By understanding how confirmation bias operates (ie, heuristic of decision making),[13] we may develop strategies to mitigate its effects and promote more accurate and effective decision making in sport science (eg, performance).

Latinjak https://orcid.org/0000-0003-4412-872X
Bertollo https://orcid.org/0000-0002-0972-9178
Boullosa https://orcid.org/0000-0002-8477-127X
Beato (M.Beato@uos.ac.uk) is corresponding author, https://orcid.org/0000-0001-5373-2211

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Understanding Confirmation Bias

Confirmation bias is a cognitive, emotional, and motivational phenomenon that leads to a perceptual distortion thus favoring information that confirms personal beliefs.[2,14] Specifically, heuristics represent cognitive shortcuts that humans employ to arrive at rapid decisions. These mental strategies are inherent to our brain's evolution and can operate either consciously or unconsciously to save time and energy.[13] While heuristics prove beneficial in situations requiring swift judgment, like making snap decisions (eg, pass the ball during a game), they can also result in suboptimal choices when more time is available (eg, strategic decisions about training) and in a context requiring processing lots of information with complexity constraints. Confirmation bias manifests in 3 primary ways: biased search for information, biased interpretation of information, and biased recall of information.

Biased Search for Information

Biased search for information occurs when individuals actively seek out information that supports their existing beliefs while ignoring or undervaluing information that contradicts them (eg, "cherry picking" their favorite source of information).

Example in Coaching

A coach who firmly believes that a particular training method, for instance, steady-state "cardio" is superior, may only look for studies and data that confirm this belief. They might search for articles (a path to hyper specialization and to partial blindness to other areas in a given field) and testimonials that highlight the benefits of this particular training modality while disregarding or dismissing evidence that suggests other training methods, like high-intensity interval training, strength training, or their combination, which could be more effective for certain athletes.

Impact on Decision Making

This selective search for information can lead to a narrow understanding of training methodologies and hinder the coach's ability to make well-rounded decisions. By focusing only on information that supports their preconceptions, the coach may miss out on valuable insights and alternative approaches that could benefit their athletes. Moreover, decisions are influenced by coach's ability and styles directly predisposing them to choose a preferred method over other methods that the coach feels less comfortable with.

These biases can lead to flawed decisions, affecting athletes' performance, training regimens, and coaching strategies. Table 1 reports some mitigations strategies for the biases reported above.

Biased Interpretation of Information

Biased interpretation of information occurs when individuals interpret ambiguous or complex information in a way that aligns with their preconceptions.

Example in Athlete Performance

An athlete who has been following a specific training regimen might attribute a successful performance solely to that regimen, even if other factors, such as improved nutrition, mental conditioning, or favorable competition conditions, played a significant role. Similarly, a coach might interpret an athlete's improvement in performance as evidence that their coaching methods are effective, without considering other contributing factors or ignoring some variables at play, including luck.

Impact on Decision Making

This biased interpretation can lead to overconfidence in certain training methods or strategies and an underestimation of the importance of other factors. It can also result in a failure to recognize the need for adjustments or changes in training programs, methodologies, and approaches.

This bias can lead to misinterpretations and flawed conclusions, particularly in the context of sports science, where data and outcomes are often multifaceted and influenced by numerous variables. To overcome this situation a multidimensional approach (ie, integrating biological, psychological, and social monitoring methods) should be taken and Table 2 reports some mitigations strategies for the biases reported above.

Biased Recall of Information

Biased recall of information involves remembering information that supports one's beliefs more readily than information that contradicts them.

Example in Coaching Strategies

A coach might recall instances where their preferred strategy led to success while forgetting or downplaying instances where it did not.

Table 1 Biased Search for Information: Mitigation Strategies

Strategy	Description
Encourage diverse perspectives	Coaches and practitioners should actively seek out and consider diverse perspectives and contradictory evidence. This can be achieved by engaging with a wide range of research, attending conferences, and participating in discussions with colleagues who hold different viewpoints. Exploring different theoretical approaches and models is recommended. Learning from other professionals.
Structured literature reviews	Implementing structured literature reviews that include objective and reproducible criteria for evaluating the quality and relevance of studies can help ensure a more balanced and comprehensive understanding of the evidence. Coaches that do not have this advanced know-how can ask for support from researchers and universities. However, practitioners need to be aware of the potential limitations of reviews and so they need to critically verify the included studies or adapt the conclusions to the specific context of their athletes/teams[15,16] while interacting with other experts and sport scientists.[17]
Critical thinking training	This type of training provides coaches and practitioners with the skills to analyze information objectively, make well-informed decisions, and apply scientific principles to their practice. Critical thinking can help to identify and challenge assumptions, evaluate evidence, improve problem-solving skills, and promote continuous learning.

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For example, a coach who believes in the effectiveness of a particular defensive or offensive strategy might remember games where this strategy worked well while forgetting games where it failed. This selective recall can reinforce the coach's belief in the strategy, even if the overall evidence suggests it is not consistently effective in all possible scenarios.

Impact on Decision Making

Biased recall can lead to an overestimation of the effectiveness of certain strategies and an underestimation of the need for change or adaptation. It can also result in a lack of learning from past mistakes and missed opportunities for improvement. This bias can lead to a skewed perception of past events and experiences, influencing future decisions and actions. Table 3 reports some mitigations strategies for the biases reported below.

Confirmation Bias in the Assessment

In sports science, measurements and assessments are essential for evaluating athletes' abilities, progress, and potential. However, confirmation bias can influence the design, interpretation, and application of the right instruments and tests.

Test Design

Researchers and practitioners may design tests that are more likely to produce results that confirm their hypotheses. For example, a strength test might be designed in a way that favors certain performance parameters over others, leading to biased conclusions about an athlete's overall strength. This type of bias could be also found in the design of research studies, where researchers would prefer that their study demonstrates their hypothesis.

Data Interpretation

When interpreting test results, confirmation bias can lead to selective attention to data that supports preexisting beliefs. For instance, a coach might focus on improvements in specific metrics while ignoring declines in others, leading to an incomplete understanding of an athlete's performance profile.

Application of Results

Confirmation bias can also affect how test results are applied in practice. A coach might use test results to justify continuing with a particular training program, even if the overall evidence suggests that a different approach would be more effective.

Practical Example: Maximum Oxygen Consumption Testing

Maximum oxygen consumption (VO_2max) tests are commonly used to measure an athlete's aerobic power. However, confirmation bias can influence both the administration and interpretation of these tests. A coach who believes that a specific aerobic training modality is the best way to improve VO_2max might design a test protocol and select exhaustion criteria to favor the observation of changes following their favorite training protocol. Additionally, when interpreting the results, the coach might only focus on improvements in VO_2max while ignoring other important factors such as the velocity or power associated to VO_2max, the anaerobic threshold or other sport-specific performance improvements associated to durability.

Practical Example: Strength Training

Strength training is a critical component of athletic development, but confirmation bias can lead to suboptimal training practices and

Table 2 Biased Interpretation: Mitigation Strategies

Strategy	Description
Use of validated objective and subjective metrics	Relying on objective and subjective validated metrics and standardized assessments can help reduce the influence of subjective interpretations. For example, using performance data, physiological measurements, and biomechanical analyses can provide a more accurate picture of an athlete's progress throughout a fine-tuning approach.
Peer review and collaboration	Engaging in peer review and collaborative decision making can help identify and correct biased interpretations. By seeking input from colleagues and experts, coaches and practitioners can gain a more balanced perspective.
Reflective practice	Encouraging reflective practice, where coaches and athletes regularly review and critically evaluate their decisions and outcomes, can help identify and address biases in interpretation. Reflective practice can be enhanced with the help of external parties, such as mentors, peers, or professional consultants. For instance, consultants can regularly review sessions, critically evaluate decisions and outcomes, and offer objective feedback and identify areas of bias or misinterpretation.

Table 3 Biased Recall: Mitigation Strategies

Strategy	Description
Maintaining detailed records	Keeping detailed records of training sessions, competitions, and outcomes can help provide an objective basis for evaluating past decisions. By systematically documenting and reviewing performance data, coaches and practitioners can gain a more accurate understanding of what works and what does not.
Regular performance reviews	Conducting regular performance reviews that involve both quantitative and qualitative assessments can help ensure a more balanced recall of information. These reviews should include input from multiple sources, such as athletes, assistant coaches, and performance analysts.
Reflective practice	Implementing feedback mechanisms that encourage honest and constructive feedback from athletes and colleagues can help identify and correct biased recall. By creating an environment where feedback is valued and acted upon, coaches and practitioners can promote continuous learning and improvement.

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outcomes. Program selection and progress tracking: coaches and athletes may favor strength training programs that align with their beliefs, even if other programs are more effective.[18,19] For example, a coach who believes in the superiority of free weights over machines might exclusively use free weights,[20] disregarding evidence that machines can also be beneficial or even better under some circumstances. Moreover, when tracking progress, confirmation bias can lead to selective attention to improvements that confirm preexisting beliefs. A coach might highlight increases in some specific strength parameters (eg, 1-repetition maximum vs rate of force development vs maximum power) while downplaying or ignoring signs of overtraining or imbalances. In this case, the attribution of success is solely and wrongly placed on the intervention, whereas unsuccessful outcomes are explained by other noncontrolled external factors (probably to justify the own failure).

Another example could be related to the periodization adopted. Periodization is a systematic approach to strength training that involves systematic variations of the training stimuli over time following the competitive schedule.[21] However, confirmation bias can influence the choice and implementation of some periodization models. A coach who believes in the superiority of linear periodization might exclusively use this model, ignoring evidence that other models, such as undulating or block periodization, could be more effective for certain athletes to develop specific outcomes in different sports and performance levels. Additionally, when evaluating the outcomes, the coach might focus on short-term gains while overlooking long-term sustainability and injury prevention, as well as other performance influencing factors as nutrition or recovery strategies. Meanwhile, the current scientific evidence does not support the superiority of periodization models over non periodization,[22] which would be better referred as programming models,[23] maybe because sports periodization cannot be exposed to falsifiability testing.

Confirmation Bias in Coaching

Coaching involves making numerous decisions about training, strategy, and athlete management. Confirmation bias can significantly impact these decisions, leading to suboptimal outcomes. This might be related to the often referred as the "art" of training.

Talent Identification

Coaches may favor athletes who fit their preconceived notions (based on their experience and not evidence) of what a successful athlete looks like (eg, anthropometric, psychophysical or technical parameters), leading to biased talent identification and development. For example, a coach might prioritize height and physicality in basketball players, overlooking other important attributes such as agility and tactical intelligence. Another example could be associated with the section of specific coaching strategy and tactics. Confirmation bias can influence the development and implementation of strategies and tactics (eg, nonlinear pedagogy). A coach who believes in a particular style of play might stick to it rigidly, even when evidence suggests that a different approach would be more effective against a specific opponent.

Confirmation Bias in Coaching Philosophy

A coach's philosophy influences the type of training and strategies used by the coach and so leads to confirmation bias. For instance, a coach who believes in the importance of mental toughness might prioritize drills and exercises that test athletes' psychological resilience, even if these practices are not the most effective for improving performance. Additionally, the coach might interpret athletes' behaviors and performances through the lens of mental toughness,[24] attributing successes to this trait while ignoring other contributing factors. Another aspect of coaching is communication,[25] therefore, the way coaches provide feedback and communicate with athletes can also be affected by confirmation bias. A coach might give more positive feedback to athletes who conform to their expectations while being overly critical of those who do not, regardless of actual performance.

Mitigating Confirmation Bias in Sport Science

To mitigate the impact of confirmation bias in sports science, it is essential to adopt strategies that promote the concurrent use (ie, fine-tuning monitoring) of validated objective (eg, global navigation satellite system) and subjective measures (eg, rate of perceived exertion) as well as evidence-based decision making. In the following lines, we have suggested some strategies for mitigating confirmation bias in sports science:

Awareness and Continuous Education

Educating coaches, athletes, and practitioners about confirmation bias and its effects is the first step in mitigating its impact. By understanding how this bias operates, individuals can become more aware of their own cognitive processes and take steps to counteract them. For instance, clubs could conduct workshops or training sessions focused on confirmation bias. Use interactive activities and real-life scenarios to help participants to recognize how these biases manifest in their decision-making processes. Teach critical thinking skills that enable individuals to evaluate information more objectively. Promote the inclusion of diverse perspectives in decision-making processes.

Diverse Perspectives

Encouraging open dialogue from diverse perspectives can help to challenge preexisting beliefs, therefore reducing the influence of confirmation bias. This can be achieved through interdisciplinary collaboration, peer review, peer mentoring, and seeking input from individuals with different backgrounds and experiences. However, to embrace these strategies, practitioners and sports scientists should interact with humility, which is a prerequisite for accepting diverse perspectives.

Evidence-Based Practices and Evidence-Informed Practices

Emphasizing the importance of evidence-based practices or evidence-informed practices can help counteract confirmation bias.[26,27] This involves critically evaluating the available evidence, considering multiple sources of information, and being open to changing one's beliefs in light of new data.

Structured Decision Making

Implementing structured decision-making processes can reduce the influence of confirmation bias. This can include using standardized

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protocols for testing and assessment of athletes, setting clear criteria for evaluating performance, and regularly reviewing and updating training programs based on objective data. However, in many instances, sports scientists and strength and conditioning coaches may be working in silos and may not necessarily be part of wider group discussions. To overcome this barrier, we suggest having regular interdisciplinary meetings (with relevant stakeholders) or regularly attending specific conferences while encouraging cross-training and education among different roles, and establishing clear feedback loops where information flows both ways. Coaches and athletes should provide feedback to sports scientists and coaches, and vice versa.

Reflective Practice

Encouraging critical reflective practice can help individuals recognize and address their own biases. This involves regularly reflecting on one's decisions, seeking feedback from others, and being open to constructive criticism.

Principle of Falsifiability

This is a key concept in the philosophy of science, introduced by philosopher Karl Popper.[1] It states that for a theory to be considered scientific, it must be possible to prove it wrong through empirical evidence. Therefore, researchers and practitioners can reduce the bias in sports by ensuring that decisions and judgments are based on objective, testable criteria rather than subjective opinions or criteria. This does not mean they do not consider subjectivity, but it is seen through the lens of phenomenology to tailor idiosyncratic interventions in elite sport.[28]

Bayesian Inference

This statistical approach allows researchers to update the probability of a hypothesis as more evidence becomes available. This method combines prior knowledge (prior probability) with new data (likelihood) to form a posterior probability.[29] This ongoing process of updating enhances the accuracy of predictions and decision. Table 4 summaries some strategies for mitigating confirmation bias in sport science.

Conclusions

This paper explores the concept of confirmation bias in sport science, providing practical examples in the contexts of testing, training, and monitoring. By understanding how confirmation bias operates (ie, heuristic of decision making), we may develop strategies to mitigate its effects and promote more accurate and effective decision making in sport science (eg, performance).

Practical Applications

Confirmation bias occurs when individuals favor information that aligns with their existing beliefs while dismissing evidence that contradicts those beliefs. Confirmation bias is a cognitive, emotional, and motivational phenomenon that leads to a perceptual distortion favoring information that confirms existing personal beliefs and so can impact sports science practices and athletes' performance, affecting testing, talent identification, training, monitoring, and coaching practices. By understanding how confirmation bias operates and adopting strategies to mitigate its effects, we can promote more accurate and effective decision making in sports.

Table 4 Strategies for Mitigating Confirmation Bias in Sport Science

Strategy	Description
Awareness and education	Educate coaches, athletes, and practitioners about confirmation bias and its effects. Understanding how this bias operates helps individuals become more aware of their cognitive processes. Encourage steps to counteract bias.
Diverse perspectives	Foster open dialogue and diverse viewpoints. Challenge preexisting beliefs by seeking input from individuals with different backgrounds. Interdisciplinary collaboration and peer review.
Evidence-based practices and/or evidence-informed practices	Emphasize evidence-based and evidence informed decision making as well as critically evaluate available evidence. Confirmation bias can arise from both practice-based evidence and evidence-informed practice. Therefore, we suggest organizing discussions in which different perspectives are considered. For example, build up teams for discussion that include both experienced coaches and sports scientists. Implement regular review and reflection sessions where decisions and outcomes are evaluated, while considering multiple information sources and be open to changing beliefs based on new data. This helps in identifying any biases and allows for adjustments based on both practical insights and scientific evidence.
Structured decision making	Implement standardized protocols for tests and assessments. Set clear criteria for evaluating performance. Regularly review and update training programs using objective and subjective data.
Critical reflective practice	Regularly reflect on decisions. Seek feedback from others. Be open to constructive criticism.
Principle of falsifiability	A theory to be considered scientific, it must be possible to prove it wrong through empirical evidence. Decisions and judgments should be based on objective, testable criteria rather than subjective opinions or criteria. Making the decision-making process transparent and subjected to review can help reduce bias.
Bayesian theory and inference	Bayesian decision theory helps in making decisions under uncertainty by considering the probabilities of different outcomes and their associated costs or benefits. Practitioners can combine different sources of information, such as athlete/player's statistics, competition/game conditions, and historical data, to improve the accuracy of predictions. Update the probability of a hypothesis as more evidence becomes available.

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As decision making becomes more evidence-based rather than belief-based within sports contexts, the entire field stands to benefit from enhanced scientific rigor leading to improved athlete outcomes and advancements in knowledge. This, in turn, can lead to better outcomes for athletes, coaches, and practitioners, ultimately advancing the field of sports science.

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